

PeerTutor - Software Requirements Specification

Peer-to-Peer Micro-Tutoring Marketplace | Django Full Stack Project

Sl.No	Requirement	Functional/ Non Functional	Essential/ Desirable	Comments
R1	The system shall allow users to register by selecting one of three mandatory roles: Student, Tutor, or Admin.	Functional	Essential	Role-based access control setup.
R2	The system shall authenticate users via a unique email address and encrypted password combination.	Functional	Essential	Core security and login requirement.
R3	The system shall restrict view access to specific dashboard components based on user role.	Functional	Essential	Ensures users only see relevant actions.
R4	The system shall allow Tutors to submit a profile specifying subjects taught, hourly rate, marksheet upload, GitHub profile link, and a project showcase.	Functional	Essential	Core credibility and trust-building data.
R5	The system shall set a newly submitted Tutor profile to Pending status until reviewed.	Functional	Essential	State tracking for onboarding.
R6	The system shall provide Admin an interface to review Tutor documents and Approve/Reject the profile.	Functional	Essential	Verification workflow, ensures honesty guarantee.
R7	The system shall restrict session listing capability to Tutors with Verified status only.	Functional	Essential	Prevents unverified tutors from transacting.
R8	The system shall allow Verified Tutors to publish available time slots per subject.	Functional	Essential	Core scheduling functionality.
R9	The system shall permit Students to search and filter tutors by subject, rating, and price.	Functional	Essential	Improves discoverability of tutors.
R10	The system shall allow Students to book an available session slot from a Verified Tutor.	Functional	Essential	Core interaction for claiming a slot.
R11	The system shall process real-money payment at the time of booking via a test-mode payment gateway.	Functional	Essential	Core monetization mechanism.
R12	The system shall hold booking payment in a pending state until the session is marked complete.	Functional	Essential	Escrow-style protection for both parties.
R13	The system shall release payout to the Tutor upon session completion, minus a platform commission of 10-15%.	Functional	Essential	Core revenue model for the platform.
R14	The system shall enforce that a Tutor's time slot cannot be double-booked by more than one Student.	Functional	Essential	Prevents scheduling conflicts.
R15	The system shall allow Tutors to cancel a scheduled session, with the Student notified and refunded.	Functional	Essential	Session lifecycle management.
R16	The system shall allow Students and Tutors to submit a rating and review of each other after a completed session.	Functional	Essential	Builds trust layer and quality signal.

Sl.No	Requirement	Functional/ Non Functional	Essential/ Desirable	Comments
R17	The system shall serve role-specific dashboards: Students see booking history, Tutors see earnings and sessions, Admin sees pending verifications and commission overview.	Functional	Essential	Personalized user experience.
R18	The system shall provide an administrative panel for superusers to oversee users, sessions, and payments.	Functional	Essential	System maintenance and oversight.
R19	The system shall display each Tutor's average rating and completed-session count on their profile.	Functional	Desirable	Reinforces credibility at a glance.
R20	The system shall compute and display real-time platform metrics such as total sessions completed and active verified tutors.	Functional	Desirable	Analytics for homepage visibility.
R21	The database shall enforce strict referential integrity through foreign key constraints.	Non Functional	Essential	Ensures robust data relationships.
R22	The application shall hash all user passwords using PBKDF2/SHA-256 before storage.	Non Functional	Essential	Strict security standard.
R23	The user interface layouts shall be optimized for laptop/desktop screen widths (1024px and above).	Non Functional	Essential	Laptop-first UI design rule per project requirement.
R24	Database lookup queries for tutor search shall execute in under 2.0 seconds.	Non Functional	Essential	Performance and speed metric.
R25	The codebase shall conform to Django's Model-View-Template (MVT) pattern.	Non Functional	Essential	Software architecture constraint.
R26	Payment and payout logic shall be isolated in a dedicated module to support auditability.	Non Functional	Essential	Financial traceability and maintainability.